

[LUP-1] Gold futures vs ETF minute notional and aggressive volume

Created: 18/Aug/26 Updated: 18/Aug/26

Status:	To Do
Project:	Lupine Research
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Task	Priority:	Medium
Reporter:	Adam Balter	Assignee:	Prateek Mishra
Resolution:	Unresolved	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

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Description

Get familiar with **gold market structure** by measuring where notional and aggressive flow actually live, in futures vs ETFs.

This is a data / market-microstructure exercise, not an alpha hunt. Charts + a short write-up are the deliverable.

Data

Venue	Symbols	Table	Docs
COMEX gold futures	GC (100 oz), MGC (10 oz)	us_futures.cme_comm_sixty (also cme_comm_five if you want 5s)	CME Futures Bars
Gold ETFs	GLD, IAU	us_equities.us_equity_sixty	Equity Bars 1m

Shared bar fields you will use: Time, symbol, vwap, vwap_traded_volume, uptick_volume, downtick_volume, plus trade_size_at_ask / trade_size_at_bid (quote-rule aggression).

Always filter on symbol and date. Contract symbols have expiry suffixes (GCZ25, MGCZ25, ...). For each day, identify **front-month** as the expiry with the most volume that day; also report **all-expiry GC** and **all-expiry MGC** so you can see roll.

Notional (USD)

Do **not** compare raw contract counts or share counts. Convert everything to USD notional:

- **GC**: vwap_traded_volume * vwap * 100
- **MGC**: vwap_traded_volume * vwap * 10
- **GLD / IAU**: vwap_traded_volume * vwap

Use vwap when present; fall back to last_trade only if vwap is missing.

Core exercises (do these)

1. Minute-of-day notional profile

For a recent window (suggest **20–40 trading days**):

- Chart **total notional by minute** over the day, **futures and ETFs separately**.
- Show GC, MGC, GC+MGC, GLD, IAU, GLD+IAU.
- Overlay session markers in ET: COMEX Globex day session, NYSE RTH 09:30–16:00, London open (03:00 ET / 08:00 London), COMEX metals close (13:30 ET), NYSE close.
- Plot both (a) a few single days and (b) the **cross-day median + p25/p75** minute-of-day profile so one loud day does not dominate.

Questions to answer in the write-up:

- Where does gold actually trade? London vs NY open vs COMEX close vs NYSE close.
- How does ETF notional compare in size to COMEX notional during RTH? Overnight?
- Does MGC matter, or is it noise next to GC?

2. Aggressive volume, separately by venue and by day

Do futures and ETFs **separately**. Do **not** mix them into one series until you have looked at each.

Per minute and **rolled up to daily totals**:

- Aggressive buy $\approx$ uptick_volume (tick rule)
- Aggressive sell $\approx$ downtick_volume
- Signed aggression = uptick – downtick
- Aggressive **notional** = those volumes $\times$ price $\times$ multiplier (same as above)

Optional second definition (quote rule): trade_size_at_ask vs trade_size_at_bid. Note if tick-rule and quote-rule disagree.

Charts:

- Daily time series of aggressive buy notional, sell notional, and net signed notional — **GC/MGC one panel, GLD/IAU another**.
- One or two days with a big net print: what happened in the other venue the same minute?

Questions:

- Is a “busy gold day” busy in both venues, or only in futures?
- Does net aggressive flow in GC line up with GLD the same day, or can they diverge?

Stretch / follow-on (pick 2–3)

These are the other standard ways to get fluent in gold for research:

1. **Product map** — Write 10 lines on GC vs MGC vs GLD vs IAU (multiplier, hours, creation unit / AUM intuition). Optional: GDX/GDXJ as miner beta, not gold itself.
2. **Front-month roll** — Plot daily volume and open-interest share (if available) of the two nearest GC expiries through a roll. When does liquidity switch?
3. **Lead-lag** — 1-minute returns: does GC lead GLD around NYSE open vs around macro prints? Simple lagged correlation is enough.
4. **Basis / fair value** — Compare GC price to GLD implied gold (GLD is ~ 0.1 oz/share historically — **check the current factor**, do not assume). When does the basis blow out?
5. **Spreads** — avg_spread by minute for GLD vs a GC bid/ask proxy. When is the ETF expensive relative to futures?
6. **Event days vs quiet days** — CPI / FOMC / NFP vs a dead Tuesday. Same charts as core exercises, two-day contrast.
7. **Closing auction** — GLD/IAU in us_auctions.arca_auc_close vs continuous volume in the last 10 minutes. Auction is a different animal than the 1m tape.
8. **GC vs MGC mix** — MGC share of COMEX gold notional by time of day (retail / small-ticket vs the big contract).

Deliverable

- Notebook or script + charts (PNG/HTML is fine).
- Half-page notes: what surprised you, where liquidity lives, and one thing you would not want to get wrong in a gold strategy.
- Drop charts and notes on this ticket.

Notes

- Compare **notional**, not lots vs shares.
- Filter startsWith(symbol, 'GC') carefully so you do not mix in unrelated roots; MGC is its own root.
- ETF bars are RTH; futures are a much longer session. Align clocks in America/New_York.
- Strat4aGold exists for later gold research. Do not start fitting alphas here.

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